

FRANKFORT, MI, USA

WMO: 720340

Lat: 44.626N

Lon: 86.201W

Elev: 632

StdP: 14.36

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 54818

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 3.6	(c) 8.7	(d) -4.2	(e) 4.5	(f) 9.0	(g) 0.2	(h) 5.7	(i) 11.8	(j) 24.2	(k) 22.5	(l) 21.5	(m) 24.1	(n) 7.3	(o) 310	(p) 0.513

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.0	83.6	70.0	81.4	68.6	79.3	67.6	72.9	79.9	71.4	78.0	70.1	76.4	8.7	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.7	115.6	77.0	69.4	110.7	75.4	67.9	104.9	74.6	36.9	80.1	35.6	78.6	34.4	76.2	77.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.1	17.8	16.1	DB	-2.3	88.0	6.4	2.1	-6.9	89.5	-10.6	90.7	-14.2	91.9	-18.8	93.4
			WB	-2.8	75.4	6.5	1.5	-7.5	76.5	-11.3	77.4	-14.9	78.2	-19.6	79.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	46.8	24.9	24.7	32.4	43.3	54.1	62.9	67.8	67.4	61.6	50.5	39.6	30.5
	DBStd	17.53	9.00	9.12	10.84	8.72	8.60	6.43	5.84	5.04	7.35	8.27	8.07	7.78
	HDD50	3369	777	709	558	240	54	1	0	0	3	96	327	604
	HDD65	7017	1242	1129	1012	652	356	115	40	33	150	456	763	1069
	CDD50	2189	0	0	12	39	180	387	553	540	352	112	14	0
	CDD65	364	0	0	1	1	18	51	128	108	50	7	0	0
	CDH74	2398	0	0	3	4	143	368	908	646	299	27	0	0
	CDH80	337	0	0	0	0	20	50	154	70	43	0	0	0
Wind	WSAvg	6.5	8.2	7.7	6.5	6.8	5.8	5.0	4.7	4.7	5.5	6.9	8.0	8.1
Precipitation	PrecAvg	34.3	2.3	1.6	2.0	2.8	3.3	3.1	3.1	3.3	3.9	3.5	2.9	2.5
	PrecMax	41.3	4.6	3.0	5.3	5.1	6.4	8.3	7.2	7.2	8.2	7.8	6.1	6.9
	PrecMin	22.7	0.6	0.5	0.4	0.8	0.9	0.7	0.7	0.9	0.8	0.9	0.8	0.6
	PrecStd	4.8	1.1	0.6	1.1	1.2	1.4	1.9	1.5	1.4	2.0	1.7	1.4	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.4	51.8	70.4	72.9	82.4	84.1	87.9	84.5	84.3	77.4	65.9	56.7
		MCWB	44.7	45.1	59.9	56.5	66.1	69.3	72.1	70.5	69.1	66.7	56.3	53.7
	2%	DB	43.6	45.4	61.1	67.9	78.6	81.4	83.9	81.8	81.0	73.2	60.2	46.4
		MCWB	40.4	41.1	53.7	53.3	65.3	67.0	70.6	69.3	69.2	63.8	52.5	43.9
	5%	DB	39.7	39.9	53.7	63.1	73.3	78.7	81.4	79.9	77.5	68.6	55.0	44.2
		MCWB	37.0	37.1	46.3	50.6	61.1	65.9	69.3	68.5	66.9	60.7	49.7	41.4
	10%	DB	37.0	36.8	46.8	58.1	69.4	75.2	79.1	77.5	74.3	63.8	52.0	41.0
		MCWB	35.1	33.8	40.7	47.7	58.7	64.5	68.0	66.6	65.8	57.4	46.9	38.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.6	46.8	60.5	61.1	69.9	73.7	75.5	74.2	73.8	69.3	57.5	53.6
		MCDB	47.4	49.5	69.5	70.0	79.4	82.3	83.9	80.6	80.1	75.5	62.7	56.3
	2%	WB	41.1	41.1	54.2	55.7	66.9	69.8	73.1	72.4	71.4	66.6	53.8	45.3
		MCDB	43.2	45.0	58.8	65.0	76.0	77.7	80.7	78.5	77.7	72.2	58.4	46.1
	5%	WB	37.5	37.3	46.3	52.0	63.8	67.9	71.4	71.0	69.6	62.0	50.4	41.5
		MCDB	39.3	40.0	51.0	61.5	71.7	75.7	78.3	76.9	75.2	66.6	55.2	44.0
	10%	WB	35.2	34.0	41.4	48.6	60.3	66.1	69.9	69.5	67.4	58.1	47.3	38.5
		MCDB	37.1	36.2	46.4	56.8	67.5	73.2	76.6	75.0	73.3	64.0	51.2	40.7
Mean Daily Temperature Range	5% DB	MDBR	9.7	11.2	15.1	16.7	19.3	18.8	18.0	17.4	17.8	14.6	11.2	8.6
		MCDBR	13.2	14.5	20.6	23.9	23.4	21.6	19.4	19.1	19.4	19.3	16.7	13.2
		MCWBR	11.9	12.3	14.3	14.1	13.2	11.3	9.9	10.4	10.7	13.0	12.1	12.5
	5% WB	MCDBR	12.3	13.9	19.7	21.3	21.0	17.8	15.8	15.1	15.7	16.6	14.8	13.2
		MCWBR	11.3	12.2	14.9	14.5	13.1	10.8	9.3	9.7	10.1	13.3	12.4	12.7
Clear-Sky Solar Irradiance	taub		0.289	0.300	0.324	0.361	0.389	0.403	0.401	0.394	0.378	0.344	0.313	0.285
	taud		2.459	2.403	2.414	2.353	2.347	2.345	2.365	2.382	2.414	2.484	2.508	2.497
	Ebn at Noon		266	283	290	286	280	275	274	272	268	263	254	256
	Edh at Noon		22	28	32	37	38	39	38	36	32	26	21	20
All-Sky Solar Radiation	RadAvg		350	640	1112	1470	1800	1960	1989	1685	1305	777	426	273
	RadStd		38	70	119	138	153	138	84	108	115	88	60	32

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (44 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page